

NUTRITION ASSESSMENT QUESTIONNAIRE

Healthy Eating Index, Double Burden of Malnutrition and BMI among Adults

A SOCIODEMOGRAPHIC

A1. Name: _____
A2. Age (completed years): _____ (eligible: adult >=18 years)
A3. Sex: ☐ Male [0] ☐ Female [1]
A4. Religion: ☐ Islam [0] ☐ Hinduism [1] ☐ Christian [2] ☐ Buddhism [3] ☐ Other [4]
A5. Profession / occupation: ☐ Service [0] ☐ Business [1] ☐ Agriculture [2] ☐ Labour [3] ☐ Homemaker [4] ☐ Student [5] ☐ Unemployed [6] ☐ Other [7]: _____
A6. Personal monthly income (BDT): _____ ☐ None [0] ☐ <=10k [1] ☐ 10,001-20k [2] ☐ 20,001-40k [3] ☐ >40k [4]
A7. Marital status: ☐ Married [0] ☐ Unmarried [1] ☐ Divorced/separated [2] ☐ Widowed [3]
A8. Educational background: ☐ None [0] ☐ Primary [1] ☐ SSC [2] ☐ HSC [3] ☐ Graduate [4] ☐ Postgraduate [5]
A9. Family size: _____ persons
A10. Total monthly family income (BDT): _____ ☐ <=10k [0] ☐ 10,001-20k [1] ☐ 20,001-40k [2] ☐ >40k [3]
A11. Family type: ☐ Nuclear [0] ☐ Joint/extended [1]
A12. Food insecurity (not enough food) in past 4 weeks: ☐ Absent [0] ☐ Present [1]
A13.+ Residence: ☐ Rural [0] ☐ Urban [1] ☐ Semi-urban [2] ☐ Slum [3]
A14.+ Smoking status: ☐ Never [0] ☐ Former [1] ☐ Current [2]
A15.+ If biologically applicable, currently: ☐ Neither [0] ☐ Pregnant [1] ☐ Lactating [2] ☐ N/A [9]

B ANTHROPOMETRY & BMI

Measure without shoes; record to one decimal where possible.

B1 Height	_____ cm	B2 Weight	_____ kg
B3 BMI	_____ kg/m ²	B4 Energy requirement	_____ kcal/day

C HEI - FOOD-GROUP CONSUMPTION

C1. Unintentional weight loss in past 3-6 months? ☐ No [0] ☐ Yes [1]; if yes: _____ kg
C2. Days in past 7 days ate whole fruits (fresh/frozen/canned in juice; not juice): _____/7
C3. Servings of 100% fruit juice typically consumed per day: _____ servings/day
C4. Days in past 7 days ate non-starchy vegetables (broccoli, tomato, carrot, cabbage): _____/7
C5. Dark green leafy vegetables (spinach, kale, amaranth, mustard greens)? F1: _____
C6. Cooked dried beans, lentils or peas? F1: _____
C7. Whole grains (brown rice, whole-wheat bread, oats, millet, sorghum)? F1: _____
C8. Refined grains (white rice/bread, noodles, plain biscuits)? F1: _____
C9. Dairy (milk, yoghurt, cheese, calcium-fortified plant milk)? F1: _____
C10. Lean protein (chicken, fish, eggs, tofu, low-fat meat)? F1: _____
C11. Processed/preserved meat (sausage, salami, smoked fish, canned meat)? F1: _____

D HEI - FATS, SODIUM & ADDED SUGARS

D1. Primary household cooking fat/oil: ☐ Soy [0] ☐ Mustard [1] ☐ Palm [2] ☐ Rice bran [3] ☐ Ghee/butter [4] ☐ Other [5]: _____
D2. Deep-fried foods (fried snacks/fish, pakora, chips)? F1: _____
D3. Fast food or commercially processed meals? F1: _____
D4. Adds extra salt at table or during cooking? F1: _____
D5. Sugar-sweetened beverages (soda, fruit drink, sweet tea/coffee)? F1: _____
D6. Sweets, cakes, pastries or confectionery? F1: _____
D7. Nuts, seeds or nut butters? F1: _____
D8. Fatty fish (hilsa, mackerel, sardine, salmon)? F1: _____
D9. Reads nutrition/ingredient labels before purchase? ☐ Never [0] ☐ Sometimes [1] ☐ Usually/always [2] ☐ N/A [9]

E FANTA/FAO DIETARY DIVERSITY - 24-HOUR RECALL

Food group	Bangladesh-relevant examples	No/Yes
E1 Cereals	rice, wheat/atta, maize, bread, noodles	☐ 0 ☐ 1
E2 White roots & tubers	potato, white yam/cassava, taro	☐ 0 ☐ 1
E3 Vitamin A-rich veg/tubers	pumpkin, carrot, orange sweet potato	☐ 0 ☐ 1
E4 Dark green leafy veg	spinach, amaranth, mustard/moringa leaves	☐ 0 ☐ 1
E5 Other vegetables	tomato, onion, eggplant, cabbage	☐ 0 ☐ 1
E6 Vitamin A-rich fruits	ripe mango/papaya, cantaloupe, apricot	☐ 0 ☐ 1
E7 Other fruits	banana, apple, guava, citrus, watermelon	☐ 0 ☐ 1
E8 Organ meat	liver, kidney, heart, other offal	☐ 0 ☐ 1
E9 Flesh meat	beef, mutton/goat, chicken, duck, pork	☐ 0 ☐ 1
E10 Eggs	chicken, duck, quail or other eggs	☐ 0 ☐ 1
E11 Fish & seafood	fresh/dried fish, shrimp, crab/shellfish	☐ 0 ☐ 1
E12 Legumes, nuts & seeds	dal, lentils, beans, chickpeas, peanuts, seeds	☐ 0 ☐ 1
E13 Milk & milk products	milk, yoghurt, cheese; exclude milk in tea	☐ 0 ☐ 1
E14 Oils & fats	oil, ghee, butter added/used in cooking	☐ 0 ☐ 1
E15 Sweets	sugar/honey, sweets, cake, sweet drinks	☐ 0 ☐ 1
E16 Spices/condiments/beverage ages	spices, sauces, tea/coffee, alcohol	☐ 0 ☐ 1

F MEAL PATTERNS & EATING BEHAVIOURS

- F1.** Main meals typically consumed per day: ☐ 1 [1] ☐ 2 [2] ☐ 3 [3] ☐ >=4 [4]
- F2.** Regularly eats breakfast within 2 hours of waking? ☐ No [0] ☐ Yes [1]
- F3.** Days/week skips at least one main meal: ____ /7
- F4.** Home-cooked meals vs outside food: ☐ Mostly home [0] ☐ Equal [1] ☐ Mostly outside [2]
- F5.** Access to clean, safe drinking water at home? ☐ No [0] ☐ Yes [1]
- F6.** Plain water consumed daily: ____ cups/day
- F7.** Prescribed or self-selected dietary pattern? ☐ No [0] ☐ Yes [1]: ☐ Vegetarian ☐ Low-salt ☐ Diabetic ☐ Weight-loss ☐ Other: ____
- F8.** Late-night meals/snacks after 9 PM? F2: ____
- F9.** Snacks between main meals? F2: ____

G DOUBLE BURDEN OF MALNUTRITION

- G1.** Previously diagnosed with undernutrition, protein-energy malnutrition (PEM) or wasting? Code: ____
- G2.** Ever diagnosed with micronutrient-deficiency anaemia (iron, folate or B12)? Code: ____
- G3.** Significant fatigue, weakness or poor wound healing recently? Code: ____
- G4.** Current/history of food insecurity (not enough food to eat)? Code: ____
- G5.** Adequate dietary protein (assessor estimate from recall)? ☐ Inadequate [0] ☐ Adequate [1] ☐ Unclear [9]
- G6.** Clinically diagnosed Type 2 diabetes mellitus? Code: ____
- G7.** Diagnosed hypertension (BP >=140/90 mmHg)? Code: ____
- G8.** Current/prior cardiovascular disease (heart attack, stroke, heart failure)? Code: ____
- G9.** Feels they consume more calories than needed? ☐ Never [0] ☐ Sometimes [1] ☐ Often [2] ☐ DK [9]
- G10.+** Currently takes medicine for diabetes, hypertension or lipids? ☐ No [0] ☐ Yes [1]: ____
- G11.+** Other diagnosed chronic condition affecting diet/weight? ☐ No [0] ☐ Yes [1]: ____

H PHYSICAL ACTIVITY & SEDENTARY BEHAVIOUR

- H1.** Days/week of moderate-intensity activity (brisk walking, cycling, household work): ____ /7; average ____ min/day
- H2.** Days/week of vigorous-intensity activity (running, heavy lifting, fast cycling): ____ /7; average ____ min/day
- H3.+** Typical sitting/reclining time while awake: ____ hours/day

I PSYCHOSOCIAL & MENTAL-HEALTH FACTORS

I1-I2 (past 2 weeks):

- I1.** Felt down, depressed or hopeless? Code: ____
- I2.** Felt nervous, anxious or on edge? Code: ____
- I3.** Overall stress in past month: ☐ None [0] ☐ Low [1] ☐ Moderate [2] ☐ High [3] ☐ Very high [4]
- I4.** Eats more than planned when stressed/emotionally upset? ☐ No [0] ☐ Sometimes [1] ☐ Yes [2]
- I5.** Positive relationship with food (without guilt, shame or fear)? ☐ No [0] ☐ Sometimes [1] ☐ Yes [2]
- I6.** Sleep per night: ____ hours

J NUTRITION KNOWLEDGE, ATTITUDES & SELF-EFFICACY

For J1-J5: [1] Strongly disagree; [2] Disagree; [3] Neutral; [4] Agree; [5] Strongly agree.

- J1.** I understand what a balanced diet means and which food groups should be included. Circle: 1 2 3 4 5
- J2.** I know the recommended daily servings of fruits and vegetables for an adult. Circle: 1 2 3 4 5
- J3.** I can identify foods high in saturated fat, added sugar or sodium on a food label. Circle: 1 2 3 4 5
- J4.** I understand the link between poor diet and chronic diseases such as diabetes and heart disease. Circle: 1 2 3 4 5
- J5.** Good nutrition is a priority for me compared with other aspects of my daily life. Circle: 1 2 3 4 5

CONSENT VOLUNTARY PARTICIPATION

I have been informed about the purpose of this study. I voluntarily agree to participate and understand that my information will be kept confidential and used only for research purposes.

C0. Consent given: ☐ No [0] - STOP ☐ Yes [1]

Respondent signature/thumbprint: _____ Date: _____

Interviewer name/signature: _____ Date: _____

NOTES ENUMERATOR / ASSESSOR COMMENTS
